HEKIMA GIRLS' SECONDARY SCHOOL
FORM THREE TERMINAL EXAMINATION, MAY 2026
FOOD AND HUMAN NUTRITION

TIME: 3:00HOURS
INSTRUCTIONS:
1. This paper consists of section A, B and C.
2. Attempt all questions in section A and B and C.

SECTION A (16 Marks):
Answer all questions in this sections

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number in the space provided.

   i) Which raising agent is commonly used in sponge cakes?
   A. Steam B. Air C. Yeast
   D. Baking soda E. Ammonia

   ii) Malnutrition can be defined as;
   A. Excess intake of food
   B. Insufficient intake of food
   C. Insufficient or excess intake of food
   D. Adequate intake of food.
   E. Overfeeding of food

   iii) The best source of vitamin C is _____
   A. A cucumber B. An orange C. A tomato
   D. A carrot E. A potato.

   iv) Depression, inflamed and painful nerves, irritability, weak muscle defective memory and anxiety are characteristics symptoms of deficiency of _____
   A. Thiamine B. Riboflavin C. Niacin
   D. Ascorbic acid E. Cyanocobalamin

   v) Perishable food that spoil rapidly usually contain relatively large amount of ____
   A. Carbohydrate B. Water C. Vitamins
   D. Proteins E. Minerals.

   vi) Biological food poisoning involves the presence of _____ the food
   A. Pesticides B. Insecticides C. Fungicides
   D. Virology E. Microorganisms.

   vii) Baking powder produces which gas?
   A. Oxygen B. Nitrogen
   C. Carbon dioxide D. Hydrogen E. Nitrogen

viii) Food can undergo spoilage if it _____

A. Gets damaged
B. Does not ripe well
C. Contains more fibre
D. Contains high moisture
E. Contains more soil

ix) Many people claim that pasta (from semolina) is the product of cereal. As an expert, identify a specific cereal from which it's obtained.

A. Wheat
B. Millet
C. Maize
D. Sorghum
E. Barely

x) Student in the classroom argue that wheat grain has protein. From the knowledge you have identify the kind of protein present in wheat.

A. Myosin
B. Albumen
C. Caseinogens
D. Gluten
E. Lysine

2. Match the action of food contamination in list A with the correct description in list B by writing the letter of the correct responses.

LIST A	LIST B
i. Food spoilage	A. Mostly used in baked products
ii. Food preparation	B. A process which occurs to food causing loss of moisture
iii. Yeast	C. Must be cleaned before being used to food
iv. Natural decay	D. Washing hand before and after visiting toilet
v. Causative of food infection	E. Bedbugs, Fungi, Virus
vi. prevention of food poisoning	F. Bacteria, Fungi, Virus
	G. Hands before handling food
	H. Working surfaces after food preparation
	I. Change in taste, smell, texture appearance and not good for human consumption.

SECTION B (54 marks)

Answer all questions in this section

3. (a) Classify vitamins according to their solubility.

(b) Outline two (2) importance of niacin in the body

(c) Briefly explain four (4) functions of minerals in the body

4. (a) Differentiate food poisoning from food spoilage

(b) Briefly explain three (3) means by which bacteria can be transferred to food.

(c) State three (3) kitchen hygienic rules to be followed in order to maintain food hygiene.

5. Compare the two (2) meals listed below, which were chosen by two school children from a canteen at lunch time and answer the questions that follows below the menu.

Choice 1	Choice 2
i. Whole meal bread roll	i. Potato chips
ii. Stewed meat	ii. Queen cakes
iii. Mixed fruit juice	iii. Carbonated drinks

a) Which meal has a greater variety (kinds) of nutrients? (Why?)

b) Which meal is more suitable for helping a child to grow well? (How?)

6. Your mother has the following kitchen equipment briefly explain the function of each item.

a) Measuring scale
b) Vegetable knife
c) Shallow frying pan

7. (a) Raising agents are used in baked foods products. Give the meaning of raising agents.

(b) With examples, explain types of raising agents. (Give 3 points with examples)

8. a) What is malnutrition

b) Differentiate between under nutrition and overnutririon

c) Describe the signs and symptoms of

i. Acute malnutrition with oedema

ii. Acute malnutrition without oedema (Give three point each)

SECTION C

Answer all questions only (30 marks)

9. Obesity is the one of the nutritional disorders that can lead to the occurrence of other diseases. Support this statement by explaining.

a) Three causes of obesity

b) Three diseases associated with obesity.

c) Three measure to be taken in order to prevent obesity.

10. Cereals are staple food in Tanzania. From cereals different types of flours can be made. Explain

a) Explain two 2 types of cereals (based on their structures and uses)

b) Advantages of unpolished flour over polished flour (4 points)

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MARKING GUIDE

1. 1@10 = 10 Marks

QN	i.	ii.	iii.	iv.	v.	vi.	vii.	viii.	ix.	x.
ANSWER	B	C	B	A	B	E	C	D	A	D

2. 1@6 = 6 Marks

LIST A	i.	ii.	iii.	iv.	v.	vi.
LIST B	I	C	A	B	F	D

3.
(a) Classify vitamins according to their solubility. (3 marks)
i. **Fat soluble vitamins – (1.5 marks)**
Vitamin A, Vitamin D, Vitamin E and Vitamin K
ii. Water soluble vitamins – (1.5 marks)
Vitamin B_1 (Thiamine)
Vitamin B_2 (Riboflavin)
Vitamin B_3 (Niacin)
Vitamin B_{12} (Cyanocobalamin)
Vitamin C (Ascorbic acid)

(b) **Outline two (2) importance of niacin in the body. (4 marks)**
Helps in energy production/metabolism of food
Maintains healthy nervous system
Keeps skin healthy
Aids proper digestion/appetite
Helps proper functioning of digestive system
Prevents pellagra

(c) **Briefly explain four (4) functions of minerals in the body. (4 marks)**
Formation of bones and teeth (Calcium and phosphorus)
Formation of blood/haemoglobin (Iron)
Regulation of body processes and fluid balance (Sodium and potassium)
Proper functioning of nerves and muscles
Formation of hormones and enzymes
Body growth and repair

4.
(a) **Differentiate food poisoning from food spoilage. (3 marks)**
Food poisoning – 1.5 marks
Illness caused by eating contaminated food containing harmful microorganisms or toxins.
Food spoilage – 1.5 marks
Undesirable change in food that affects taste, smell, colour or texture making it unfit for consumption.

(b) **Three (3) means by which bacteria can be transferred to food. (3 marks)**
Through dirty hands during food handling
Through coughing or sneezing over food
Through contaminated utensils/equipment
Through flies, cockroaches and rodents
Through dirty water
Through raw foods contaminating cooked foods (cross contamination)

(c) Kitchen hygienic rules to be followed in order to maintain food hygiene. (3 marks)
Wash hands before handling food
Keep kitchen utensils clean
Cover food properly
Dispose waste properly
Clean working surfaces regularly
Keep pests away from kitchen
Wash fruits and vegetables before use

5.
a) Which meal has a greater variety of nutrients? (4.5 marks)
Correct Answer

Choice 1 has a greater variety of nutrients. – 2 marks

Explanation:2.5 Marks
Whole meal bread provides carbohydrates and fibre.
Stewed meat provides proteins, fats and minerals.
Mixed fruit juice provides vitamins and water.

b) **Which meal is more suitable for helping a child to grow well? (4.5 marks)**
Correct Answer:

Choice 1 is more suitable for helping a child to grow well. – 2 marks

Explanation: 2.5 marks
Contains body-building foods such as meat (protein).
Contains protective foods from fruit juice (vitamins).
Provides balanced nutrients necessary for growth and development.

6. (3@ 3= 9 marks)
Briefly explain the function the following kitchen equipment of each item.
(a) Measuring scale
Used for weighing/measuring ingredients accurately before cooking or baking
(b) Vegetable knife
Used for peeling, trimming and cutting vegetables and fruits
(c) Shallow frying pan
Used for shallow frying foods using a small amount of oil/fat

7. (a) **Meaning of raising agents. (3 marks)**
Raising agents are substances added to flour mixtures to make them rise/increase in volume. They produce gas or air which makes baked products light and soft. Example, yeast, bicarbonate of soda
(b) 3 Types of raising agents. (2@ 3 = 6 marks)
- Biological raising agents – 1 marks
Use living organisms such as yeast to produce carbon dioxide through fermentation. – 1 mark
Example: bread, buns or doughnuts. – 1 mark
- Chemical raising agents – 1 marks
Produce carbon dioxide through chemical reaction during baking. – 1 mark
Example: baking powder or bicarbonate of soda used in cakes and scones. – 1 mark
- Mechanical/Physical raising agents (Air) – 1 marks
Air is incorporated during whisking, beating or creaming and expands during cooking. – 1 mark
Example: sponge cakes. – 1 mark
- Steam as a raising agent – 1 marks
Steam formed during heating expands and raises the mixture. – 1 mark
Example: choux pastry or puff pastry. – 1 mark

8. (a) 3 Marks
Malnutrition is a condition resulting from insufficient, excessive or imbalance intake of nutrients in the body. Example, stunting, overweight and obesity.
(b) Differentiate between undernutrition and overnutrition. (3 marks)
Marking Scheme
Undernutrition – 1.5 marks
A condition caused by inadequate intake of nutrients or food needed by the body. – 2 marks
Overnutrition – 1.5 marks
A condition caused by excessive intake of nutrients or food beyond body requirements.
(c) 3 Signs and symptoms of:
i. 3 signs: Acute malnutrition with oedema (Kwashiorkor) (1.5 marks)
Swollen body/feet/face due to oedema
Thin muscles with swollen abdomen
Hair becomes reddish, thin or falls out
Skin becomes dry/flaky
Irrability and poor appetite
Stunted growth

ii. 3 signs: Acute malnutrition without oedema (Marasmus) (1.5 marks)
Severe wasting/thin body
Prominent ribs and bones
Dry wrinkled skin
Little or no body fat
Weakness and fatigue
Sunken eyes/old man appearance

9. Introduction (2 marks)
Obesity is one of the nutritional disorders that can lead to other diseases.
(a) Three causes of obesity (3 marks)
Marking Scheme (any three)
Excessive intake of high-energy foods (fatty and sugary foods) – 1 mark
Lack of physical exercise/sedentary lifestyle – 1 mark
Overeating/poor eating habits – 1 mark
Genetic factors/heredity – 1 mark
Hormonal imbalance – 1 mark
(b) Three diseases associated with obesity (4.5 marks)
Diabetes mellitus (Type 2) – 1.5 mark
Hypertension (high blood pressure) – 1.5 mark
Heart diseases/coronary heart disease – 1.5 mark
Stroke – 1.5 mark
Arthritis/joint problems – 1.5 mark
(c) Three measures to prevent obesity (4.5 marks)
Regular physical exercise – 1.5 mark
Eating a balanced diet with less fat and sugar – 1.5 mark
Avoid overeating/control portion sizes – 1.5 mark
Increase intake of fruits and vegetables – 1.5 mark
Avoid sedentary lifestyle – 1.5 mark

10.
Cereals are grain crops that form the main staple foods in many countries including Tanzania. They are important sources of carbohydrates, and from them different types of flour are produced for various foods such as ugali, bread, and porridge. (2 marks)

a)**4 marks**

Wheat. Wheat is widely used for making bread, cakes, and pasta. A wheat grain has three main parts. The outer covering is called the **bran**, which protects the grain and contains a lot of fibre. The middle part is the **endosperm**, which is the largest part and contains starch used as energy in the body. The inner part is the **germ**, which is the embryo of the seed and contains oils, vitamins, and minerals needed for growth. (2 marks)

Maize, which is the most common staple food in Tanzania. Maize grain also has three main parts. The **bran** forms the outer layer that protects the grain. The **endosperm** is the largest part and is mainly made up of starch, which provides energy. The **germ** is the living part of the seed that contains fats, vitamins, and minerals and is responsible for germination (2 marks)
(b) 4 points - Advantages of unpolished flour over polished flour (8 marks)

- Contains more fibre which aids digestion
- Retains vitamins especially B-complex
- Contains more minerals (iron, calcium)
- More nutritious than polished flour
- Helps prevent constipation
- Has better health benefits overall

Conclusion: (1 marks) - Generally, unpolished flour is healthier and more beneficial for human consumption.